Year 12 Further Maths - Discrete random variables

Similar questions - Expectation, variance and linear coding

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The probability distribution for the discrete random variable W is given in the table.

w	1	2	3	4
$P(W = w)$	0.44	0.32	p	p^2

(a) Show that $p = 0.2$. **[3]**

(b) Show that $\text{Var}(W) = 0.7744$. **[5]**

(c) Find $\text{Var}(3W + 6)$. **[1]**

2 A spinner is equally likely to land on 1, 2, 3 or 4. In a competition the spinner is spun twice and the outcomes are added to give a total score T.

(a) Show that $E(T) = 5$. **[3]**

(b) Find the variance of T. **[3]**

A competitor pays £1.80 to enter and receives £X, where $X = 0.4T$.

(c) Find the expected profit of the competitor and the variance of the competitor's profit. **[3]**

3 The discrete random variable A has the following distribution.

a	1	2	5	10	20
$P(A = a)$	0.2	0.1	0.3	0.15	0.25

(a) Find $E(A)$. **[2]**

(b) Determine $\text{Var}(A)$. **[3]**

(c) A represents the value in pence of a coin chosen at random from a pile. Ravi picks one coin from the pile at random. He then adds another coin of the same value as the one he has chosen, and one 20p coin. Find the mean and the variance of the total value of the three coins. **[3]**

4 The discrete random variable X has the following probability distribution.

x	1	2	3	4
$P(X = x)$	q	0.46	0.3	q^2

(a) Determine the value of q. **[3]**

(b) It is given that $E(aX + b) = \text{Var}(aX + b) = 15.69$, where a and b are positive constants. Determine the value of a and the value of b. **[6]**

5 A discrete random variable W has the probability distribution shown in the table, in which a and b are constants.

w	40	41	42	43	44	45
$P(W = w)$	a	b	0.25	0.25	0.1	0.1

It is given that $E(W - 42) = 0.2$. Determine the value of $\text{Var}(3W - 10)$. **[7]**

6 The random variable W can take values 1, 2, 3 or 4 and has a discrete uniform distribution.

(a) Write down the value of $E(3W)$. **[1]**

(b) Find the value of $\text{Var}(3W)$. **[2]**

(c) Determine the value of the constant k for which $E(3W + k) = \text{Var}(3W + k)$. **[2]**

The random variable S has the probability distribution shown in the following table.

s	1	2	3	4	5
$\mathrm{P}(S=s)$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{4}$	$\frac{1}{6}$

(d) Calculate $\mathrm{Var}(S)$. **[3]**

Question 1
(a) $0.44+0.32+p+p^2=1$, so $p^2+p-0.24=0$, $(p-0.2)(p+1.2)=0$. As $p>0$, $p=0.2$
(b) $\mathrm{E}(W)=0.44+0.64+0.6+0.16=1.84$. $\mathrm{E}(W^2)=0.44+1.28+1.8+0.64=4.16$. $\mathrm{Var}(W)=4.16-1.84^2=0.7744$
(c) $9\times 0.7744=6.97$

Question 2
(a) E(one spin) $=2.5$, so $\mathrm{E}(T)=2(2.5)=5$
(b) Var(one spin) $=\dfrac{1+4+9+16}{4}-2.5^2=1.25$. Spins are independent, so $\mathrm{Var}(T)=2.5$
(c) Expected profit $=0.4(5)-1.8=$ £0.20. Var(profit) $=0.4^2(2.5)=0.4$

Question 3
(a) $\mathrm{E}(A)=0.2+0.2+1.5+1.5+5=8.4$
(b) $\mathrm{E}(A^2)=0.2+0.4+7.5+15+100=123.1$; $\mathrm{Var}(A)=123.1-8.4^2=52.54$
(c) Total $=2A+20$. Mean $=2(8.4)+20=36.8$p; variance $=4(52.54)=210.16$

Question 4
(a) $q+0.46+0.3+q^2=1$, so $q^2+q-0.24=0$ and $q=0.2$
(b) $\mathrm{E}(X)=0.2+0.92+0.9+0.16=2.18$; $\mathrm{E}(X^2)=0.2+1.84+2.7+0.64=5.38$; $\mathrm{Var}(X)=5.38-2.18^2=0.6276$.
$a^2(0.6276)=15.69$ gives $a^2=25$, $a=5$. $5(2.18)+b=15.69$, so $b=4.79$

Question 5
$a+b+0.25+0.25+0.1+0.1=1$, so $a+b=0.3$. $\mathrm{E}(W-42)=-2a-b+0.25+0.2+0.3=0.2$, so $2a+b=0.55$.
Hence $a=0.25$, $b=0.05$. Using $W-42$: $\mathrm{E}[(W-42)^2]=4(0.25)+0.05+0.25+0.4+0.9=2.6$, so $\mathrm{Var}(W)=2.6-0.2^2=2.56$. $\mathrm{Var}(3W-10)=9(2.56)=23.04$

Question 6
(a) $\mathrm{E}(W)=2.5$, so $\mathrm{E}(3W)=7.5$
(b) $\mathrm{Var}(W)=1.25$, so $\mathrm{Var}(3W)=11.25$
(c) $7.5+k=11.25$, so $k=3.75$
(d) $\mathrm{E}(S)=3$ by symmetry; $\mathrm{E}(S^2)=\frac{1}{6}+1+\frac{3}{2}+4+\frac{25}{6}=10.8333$; $\mathrm{Var}(S)=10.8333-9=\frac{11}{6}=1.83$